ARYAN MEENA

Data Scientist | Python, SQL, PyTorch, Machine Learning
araj7042@gmail.com | (857) 425-6779 | LinkedIn | GitHub | Portfolio | Boston, MA (open to relocation)

EDUCATION

Master of Science in Applied Data Analytics - Boston University | GPA 3.75/4.0 **May 2026**
Bachelor of Technology in Electronics and Communication Engineering - Indian Institute of Space Science and Technology **May 2023**
Relevant Coursework: Advanced Machine Learning, Foundations of Machine Learning, Big Data Analytics, Data Mining
Honors: First runner-up of 40+ teams, BU MedAI Hackathon (team lead); 3D PET model at validation MAE 7.31 Centiloid (r 0.968)

PROFESSIONAL EXPERIENCE

Research Assistant, Data Science and Quantitative Finance, Boston University | *Boston, MA* **Sep 2025 – May 2026**
Skills – Python (pandas, NumPy), SQL, scikit-learn, time series, statistical testing, Git

- Traced a 15-year Dollar Index trading signal to bad opening prices in a free data feed by checking the feed's data quality against exchange data in Python. **Published as first author in Risks (MDPI).**
- Forecast 5-day currency volatility with time series and machine learning models at out-of-sample R^2 0.25 vs a 0.05 noise ceiling, where a 7-variable regression beat the benchmark by more than a random forest.
- Caught 5 feature engineering bugs before model fitting, one of which would have leaked 615 false zeros into the data.
- Rewrote the NumPy backtest engine behind 1,312 strategy configurations to run 17x faster with identical results, and kept every table reproducible with one command under Git version control.
- Wrote the SQL and Python behind a live research dashboard on a 12-table relational database (star schema, 7 views, 484K monthly rows) so non-specialists can check every result without code.

Graduate Teaching Assistant, Boston University | *Boston, MA* **Jan 2026 – May 2026**
Skills – Python, scikit-learn, PyTorch, statistics

- Held weekly office hours for 60+ graduate students training as data scientists in Advanced Machine Learning and Data Science with Python, debugging student code and explaining model results in plain language.
- Graded classification, regression, clustering and deep learning projects to one standard across the cohort, with written feedback on each model and analysis.

Machine Learning Research Intern, ISRO Space Applications Centre | *Ahmedabad, India* **Jun 2022 – Jul 2022**
Skills – PyTorch, U-Net, DeepLabV3+, QGIS

- Trained U-Net and DeepLabV3+ deep learning models in PyTorch to segment roads in 6,723 aerial images, and traced a drop in accuracy on Indian satellite imagery to a sensor resolution and metadata mismatch.

PROJECTS

Hospital Readmission Risk Model | *Python, LightGBM, PyTorch, Tableau* **Live dashboard**

- **Beat a published 30-day readmission benchmark (AUC 0.689 vs 0.664)** with a clinical prediction model on 101,766 electronic health record (EHR) encounters, using gradient-boosted trees and patient-history features.
- Ran an algorithmic fairness audit by age, race and sex (no race or sex gap, stable calibration), recommended age-specific thresholds after AUC fell from 0.733 to 0.631 across age bands, and wrote TRIPOD+AI model documentation.
- Published a care-team dashboard in Tableau where the top 5% of flagged patients had 3x the readmissions of a random 5%.

JudgeGuard: LLM Evaluation | *Python, FastAPI, Docker, Hugging Face* **Live app**

- Tested a small open-source large language model (Qwen3-4B) as an LLM-as-a-judge for generative AI answers. The model picked the answer supported by the source in **96 of 100 test pairs** (95% CI 92–99%).
- Deployed the judge as a REST API (FastAPI) in Docker and a live Hugging Face app, with 150 automated tests in GitHub Actions CI.

WikiFlow: Editor Dropout Prediction | *PySpark, Google Cloud Dataproc, BigQuery* **Springer paper**

- Processed **60M Wikipedia edit events** (~25 GB) through a Spark data pipeline on a 5-node Google Cloud cluster to predict which of 1.16M new editors drop out, in about 2 hours for roughly $2.50.
- Reached AUC 0.909 (precision and recall 0.87) with a distributed neural network, 6.4 points above logistic regression.

PUBLICATIONS

Meena and Pinsky. Two Blind Spots in Backtest Validation. *Risks (MDPI), 14(11), 247, 2026.*
Meena et al. Accelerated CNN Training with Genetic Algorithm Optimization. *IEEE IATMSI, 2024.*

TECHNICAL SKILLS

Programming and ML: Python (pandas, NumPy, scikit-learn), SQL, R, PyTorch, LightGBM, XGBoost, deep learning, NLP, LLM evaluation
Statistics: regression, hypothesis testing, bootstrap, cross-validation, calibration, time series forecasting
Data, Cloud and Tools: Spark (PySpark), PostgreSQL, AWS (S3, Glue, Athena), GCP (BigQuery), Tableau, Power BI, Git, Docker, CI/CD